Journal of Sound and Vibration 561 (2023) 117797

Contents lists available at ScienceDirect

Journal of Sound and Vibration

journal homepage: www.elsevier.com/locate/jsvi

Study of coupled waves of cylinder walls and internal liquid based on cylindrical shell theory and wave equation

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ARTICLE INFO

Keywords:
Coupled wave
Flow-induced vibration
Shell theory
Wave equation
Fluid added mass

ABSTRACT

In the present study, we focus on coupled waves of cylinder walls and an internal liquid, one of the important issues in various fluid machinery including nuclear reactors. The main objective of this study is to propose a novel method to predict coupled wave frequencies based on the cylindrical shell theory and wave equation with the consideration of the fluid added mass. First, we introduce the fluid added mass to consider the effects of the liquid mass. Second, we discuss the vibration behavior of cylinder walls and sound propagation of an internal liquid separately to describe the coupled wave theoretically. In this discussion, we derive dispersion relations of the cylinder walls and internal liquid based on the Flügge's cylindrical shell theory and wave equation in the cylindrical coordinates, respectively. To validate our proposed theory, we conduct an experiment on the coupled waves using a SUS304 pipe as a cylinder and water as an internal liquid. As a result of frequency analysis based on the power spectral density (PSD), we confirm that the coupled waves occur without any external vibration sources, and the vibration modes and the most prominent vibration mode vary with the flow rate of the internal water. Also, theoretical values of coupled wave frequencies consistently agree with experimental values. This result demonstrates that coupled wave frequencies can be predicted correctly using dispersion relations of cylinder walls and an internal liquid derived based on the cylindrical shell theory and wave equation with the consideration of the fluid added mass. The prediction of coupled wave frequencies for liquid-filled structures by our proposed method leads to the prevention of significant levels of vibrations and noises by making a difference between coupled wave frequencies and mechanical vibration frequencies.

1. Introduction

The flow-induced vibration is one of the important issues in various fluid machinery. This is caused by the contact of high energy flow to flexible structures. Particularly in nuclear reactors, this vibration is caused by the reactor coolant impinging on flexible reactor internals, fuel rods, shielding materials, or heat transfer tubes [1]. The resultant vibrations of structures and fluids are often undesirable and unexpected, which can cause fatigue or failure of structures. Also, acoustic detection systems attract more attention in nuclear reactors in terms of their advantage of a short time response [2–4]. The acoustic methods are divided into two

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https://doi.org/10.1016/j.jsv.2023.117797
Received 30 October 2022; Received in revised form 20 April 2023; Accepted 17 May 2023
Available online 20 May 2023
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Nomenclature

A_{nml}	Amplitude of the velocity potential of the cylindrical sound field
A_r	Amplitude of the cylindrical displacement in the radial direction
A_z	Amplitude of the cylindrical displacement in the axial direction
A_θ	Amplitude of the cylindrical displacement in the circumferential direction
c	Speed of sound
d_{in}	Inner diameter of the cylinder
d_{mean}	Mean diameter of the cylinder $d_{\mathrm{mean}} = (d_{\mathrm{out}} + d_{\mathrm{in}})/2 = 2R$
d_{out}	Outer diameter of the cylinder
E	Young's modulus of the cylinder
f	Frequency
F	Sampling rate
h	Thickness of the cylinder $h = (d_{\mathrm{out}} - d_{\mathrm{in}})/2$
j	Imaginary unit
J_n	nth order Bessel function of the first kind
k	Nondimensional thickness parameter $k = h^2/12R^2 = h^2/3{d_{\mathrm{mean}}}^2$
k_z	Axial wave number
K_0, K_1, K_2	Constants of the equation for the natural frequencies of the cylinder vibration
L	Number of segments in Welch's method
$[\mathcal{L}_{\mathrm{D-M}}]$	Matrix differential operator according to the Donnell–Mushtari theory
$[\mathcal{L}_{\mathrm{MOD}}]$	Modifying matrix differential operator
m_{c}	Mass of the cylinder per unit length
m_{f}	Mass of the fluid per unit length (fluid added mass per unit length)
M	Number of data points of each segment in Welch's method
$[\mathcal{M}_{\mathrm{D-M}}]$	Coefficient matrix according to the Donnell–Mushtari theory
$[\mathcal{M}_{\mathrm{MOD}}]$	Modifying coefficient matrix
n	Circumferential vibration mode
$\boldsymbol{n}$	Outward normal vector at the boundary of the cylinder
N	Number of data points of discrete original signals in Welch's method
r	Radial variable in the cylindrical coordinates
$\boldsymbol{r}$	Position vector
R	Mean radius of the cylinder $R = d_{\mathrm{mean}}/2 = (d_{\mathrm{out}} + d_{\mathrm{in}})/4$
s	Nondimensional length parameter $s = z/R$
S	Number of data points to shift between segments in Welch's method
t	Time
u_{nm}	mth solution of $\partial J_n/\partial x = 0$
u_r	Cylindrical displacement in the radial direction
u_z	Cylindrical displacement in the axial direction
u_θ	Cylindrical displacement in the circumferential direction
w	Window function in Welch's method
W	Power parameter factor of the window function in Welch's method
x	Discrete original signals in Welch's method
X_l	lth windowed discrete Fourier transform in Welch's method
z	Axial variable in the cylindrical coordinates
α_{nml}	Phase angle of the velocity potential of the cylindrical sound field
Γ	Gamma function
$\Delta K_0, \Delta K_1, \Delta K_2$	Modifying constants of the equation for the natural frequencies of the cylinder vibration
θ	Circumferential variable in the cylindrical coordinates
λ	Nondimensional frequency parameter $\lambda^2 = \rho_{\mathrm{all}}(1-\nu^2){d_{\mathrm{mean}}}^2\omega^2/(4E)$
ν	Poisson's ratio of the cylinder
ν_{DOF}	Degrees of freedom of χ^2 distribution $\nu_{\mathrm{DOF}} = 2L$

ξ	Nondimensional length parameter $\xi = k_z d_{\mathrm{mean}}/2$
ρ_{all}	Density of the cylinder with the consideration of the fluid added mass
ρ_{c}	Density of the cylinder
ρ_{f}	Density of the fluid
ϕ	Velocity potential of the cylindrical sound field
χ^2	χ^2 distribution
ω	Angular frequency

types: active and passive methods. In active methods, sensors on receiving sides measure the ultrasonic from sensors on sending sides in most cases. As for passive methods, sensors measure sound signals from physical phenomena. In this property, passive methods have issues with the removal and separation of background noises. Therefore, it is necessary to identify the sound and vibration that can be background noises including the sound of liquids flowing inside pipes assumed in heat transfer tubes. For these reasons, it is important to analyze the sound propagation and vibration behavior of structures with fluids inside correctly based on the theory and prevent flow-induced vibrations.

The coupled wave, one type of flow-induced vibrations, consists of the vibration of structures and the sound wave of internal or external fluids. This phenomenon is relatively complex because of the necessity to consider both the vibration behavior of structures and the wave propagation of fluids. To this problem with structural and acoustical coupling, various methods are proposed such as analytical modal descriptions of coupled waves, finite differences, finite elements, and boundary elements [5]. Also, some studies focus on the coupled wave in thin-walled cylindrical shells with fluids inside, which can be seen in various engineering structures such as pipes and ducts [6–9]. These pipe or duct systems can be vibrationally excited by a lot of sources related to the structural or internal fluid path through the drive of compressors, pumps, and valves. The more important fact is that the propagation of vibration energy along pipe systems can excite other equipment attached to structures [8]. In these systems, sound waves in the high frequency band propagate axially in the cylinder not only as plane waves but also as higher order mode waves with radial and circumferential wave bellies. When the coupled wave frequencies agree with those of vibrations and sound waves induced by blowers and pumps, the vibrations and noises can reach significant levels. Therefore, the coupled waves of cylinder walls and an internal fluid are needed to be considered in detail for the reduction of vibrations and noises [9].

One of the methods to describe the coupled waves is to consider the vibration of structures and the sound propagation of an internal fluid separately. First, the cylindrical shell theory is widely used to describe the vibration behavior of cylinder walls. In general, different eighth order systems of equations are used for modeling. In this case, the vibration behavior is described as the product of the matrix differential operator and the displacement vector [10]. This matrix differential operator depends on each shell theory; many theories are proposed by Donnell [11], Love [12], Timoschenko [13], Flügge [14,15], and so on. A comprehensive summary and discussion of these theories were already conducted by Leissa [10], and a semi-analytical procedure was discussed in detail by Farshidianfar [16]. Second, the sound propagation of the internal fluid is described by the wave equation. Assuming that the sound wave is the time-harmonic oscillation, the Helmholtz equation is derived from the wave equation. In the cylindrical coordinates, the solution of the Helmholtz equation is obtained by the separation of axial, circumferential, and radial variables. Based on the theory mentioned above, Murakami et al. investigated coupled waves of cylinder walls and an internal gas [9]. In this previous study, the authors confirmed that the coupled waves by higher order mode waves occurred at specific frequencies only under the condition where cylinder walls and internal air were vibrationally excited simultaneously by a jet flow sound as a vibration source. Also, they showed that these frequencies could be predicted by dispersion relations of the cylinder walls and internal air, each of which was derived from the Donnell–Mushtari's cylindrical shell theory and wave equation. The prediction of coupled wave frequencies by this method has the possibility to prevent significant vibrations or noises which can lead to serious accidents. Moreover, Shioyama et al. showed that coupled waves of cylinder walls and an internal gas occurred in the experimental system with an air jet flow and a pressure difference by a flow control valve [17]. On the other hand, when an internal fluid is liquid, the vibration and sound propagation behavior can be different from that of gas due to the viscosity or density of the internal fluid [18]; the viscosity and density of liquids are usually bigger than those of gas. However, little is known about this case, and no method has been proposed to predict coupled wave frequencies of cylinder walls and an internal liquid.

Based on the above, we focus on the coupled waves of cylinder walls and an internal liquid. The objectives of this study are to investigate coupled waves including higher order modes and to propose a novel method to predict the coupled wave frequencies based on the cylindrical shell theory and wave equation with the consideration of the fluid added mass. The prediction of coupled wave frequencies for liquid-filled structures by our proposed method makes it possible to prevent significant levels of vibrations and noises by making a difference between coupled wave frequencies and mechanical vibration frequencies. This paper begins with the theory of coupled waves. In this phase, we first introduce the fluid added mass to consider the mass of the internal liquid. Second, dispersion relations of the cylinder walls and internal liquid are derived based on the Flügge's cylindrical shell theory and the wave equation in the cylindrical coordinates, and the possible coupled wave frequencies are calculated. This is followed by the description of the experimental arrangement to validate our proposed theory. In this experiment, we use a SUS304 pipe as a cylinder and water as an internal liquid. Finally, we analyze the coupled waves based on the power spectral density (PSD) and discuss the validity of our proposed method to predict the coupled wave frequencies.

2. Theory of coupled waves

2.1. Fluid added mass

Whenever acceleration is imposed on a body with liquid inside, additional inertial forces act on the body due to the added mass effect [19]. For this reason, it is necessary to consider the fluid added mass in representing the vibration behavior of the body with liquid inside as in previous studies [20,21]. In this study, by adding the fluid added mass to contact parts between the cylinder and the fluid in an equal distribution, we solve the vibration behavior of the cylinder with liquid inside similarly to the case of the cylinder in the gas. When fluid with a density ρ_f exists in a cylinder with a density ρ_c, an inner diameter d_{in}, and an outer diameter d_{out}, the fluid added mass per unit length is expressed by the following equation [22]:

$$m_f = \rho_f \left\{ \pi \left(\frac{d_{in}}{2} \right)^2 \right\} \tag{1}$$

The mass of the cylinder per unit length is calculated by the following equation:

$$m_c = \rho_c \left\{ \pi \left(\frac{d_{out}}{2} \right)^2 - \pi \left(\frac{d_{in}}{2} \right)^2 \right\} \tag{2}$$

Using Eqs. (1) and (2), the density of the cylinder including the fluid added mass is expressed by the following equation:

$$\begin{aligned} \rho_{all} &= \frac{m_c + m_f}{\pi \left(\frac{d_{out}}{2} \right)^2 - \pi \left(\frac{d_{in}}{2} \right)^2} \\ &= \frac{\rho_c \left\{ \pi \left(\frac{d_{out}}{2} \right)^2 - \pi \left(\frac{d_{in}}{2} \right)^2 \right\} + \rho_f \left\{ \pi \left(\frac{d_{in}}{2} \right)^2 \right\}}{\pi \left(\frac{d_{out}}{2} \right)^2 - \pi \left(\frac{d_{in}}{2} \right)^2} \end{aligned} \tag{3}$$

The value of ρ_{all} in Eq. (3) is used as the density of the cylinder with liquid inside.

2.2. Dispersion relations of cylinder walls

In this section, dispersion relations of cylinder walls are derived based on the cylindrical shell theory [10]. First, the cylindrical coordinates are introduced as shown in Fig. 1. Also, an axial variable z is replaced by a nondimensional length s:

$$s = \frac{z}{R} \tag{4}$$

where R is the mean radius of the cylinder ($R = d_{mean}/2 = (d_{out} + d_{in})/4$). To model the vibration behavior of cylinder walls, different eighth order systems of equations are commonly used, and they can be written in matrix forms as follows:

$$\left([\mathcal{L}_{D-M}] + k[\mathcal{L}_{MOD}] \right) \begin{bmatrix} u_z \\ u_\theta \\ u_r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{5}$$

where $[\mathcal{L}_{D-M}]$ is the matrix differential operator according to the Donnell–Mushtari theory, $[\mathcal{L}_{MOD}]$ is a modifying operator which extends the Donnell–Mushtari theory to other theories. Also, u_z, u_θ, and u_r are displacements in the axial, circumferential, and radial directions, respectively, and k is the nondimensional thickness parameter:

$$k = \frac{h^2}{12R^2} \tag{6}$$

where h is the thickness of the cylinder. In the present study, we introduce the Flügge's shell theory, which gives relatively exact solutions for thin-walled cylinders [23,24]. Each component of $[\mathcal{L}_{D-M}]$ and $[\mathcal{L}_{MOD}]$ is described as follows:

$$\begin{aligned} [\mathcal{L}_{D-M}]_{11} &= \frac{\partial^2}{\partial s^2} + \frac{(1-\nu)}{2} \frac{\partial^2}{\partial \theta^2} - \frac{\rho_{all}(1-\nu^2)R^2}{E} \frac{\partial^2}{\partial t^2} \\ [\mathcal{L}_{D-M}]_{22} &= \frac{(1-\nu)}{2} \frac{\partial^2}{\partial s^2} + \frac{\partial^2}{\partial \theta^2} - \frac{\rho_{all}(1-\nu^2)R^2}{E} \frac{\partial^2}{\partial t^2} \\ [\mathcal{L}_{D-M}]_{33} &= 1 + k \left(\frac{\partial^2}{\partial s^2} + \frac{\partial^2}{\partial \theta^2} \right)^2 + \frac{\rho_{all}(1-\nu^2)R^2}{E} \frac{\partial^2}{\partial t^2} \\ [\mathcal{L}_{D-M}]_{12} &= [\mathcal{L}_{D-M}]_{21} = \frac{(1+\nu)}{2} \frac{\partial^2}{\partial s \partial \theta} \\ [\mathcal{L}_{D-M}]_{13} &= [\mathcal{L}_{D-M}]_{31} = \nu \frac{\partial}{\partial s} \\ [\mathcal{L}_{D-M}]_{23} &= [\mathcal{L}_{D-M}]_{32} = \frac{\partial}{\partial \theta} \end{aligned} \tag{7}$$

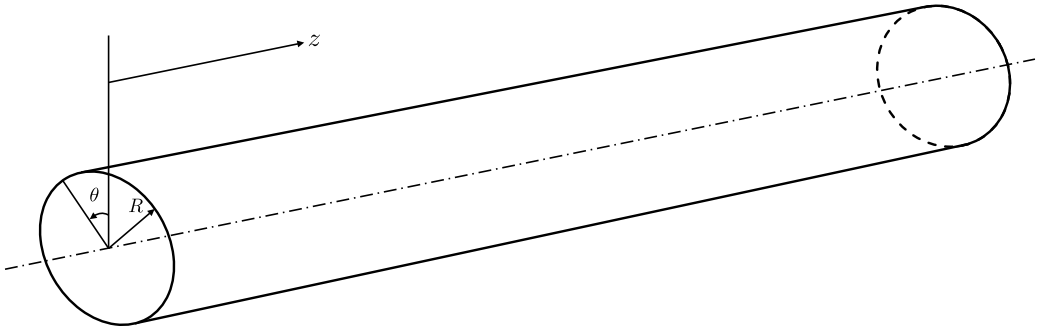


Fig. 1. Cylindrical shell and coordinate system.

and

$$
\begin{aligned}
[\mathcal{L}_{\mathrm{MOD}}]_{11} &= \frac{(1-\nu)}{2}\frac{\partial^2}{\partial\theta^2} \\
[\mathcal{L}_{\mathrm{MOD}}]_{22} &= \frac{3(1-\nu)}{2}\frac{\partial^2}{\partial s^2} \\
[\mathcal{L}_{\mathrm{MOD}}]_{33} &= 1 + 2\frac{\partial^2}{\partial\theta^2} \\
[\mathcal{L}_{\mathrm{MOD}}]_{12} &= [\mathcal{L}_{\mathrm{MOD}}]_{21} = 0 \\
[\mathcal{L}_{\mathrm{MOD}}]_{13} &= [\mathcal{L}_{\mathrm{MOD}}]_{31} = -\frac{\partial^3}{\partial s^3} + \frac{(1-\nu)}{2}\frac{\partial^3}{\partial s\partial\theta^2} \\
[\mathcal{L}_{\mathrm{MOD}}]_{23} &= [\mathcal{L}_{\mathrm{MOD}}]_{32} = -\frac{(3-\nu)}{2}\frac{\partial^3}{\partial s^2\partial\theta}
\end{aligned}
\tag{8}
$$

where E and ν denote the Young's modulus and the Poisson's ratio of the cylinder, respectively. We assume that the equations of motion of cylinder walls Eq. (5) have displacements with separability of time and space as follows:

$$
\begin{aligned}
u_z &= A_z \cos\xi s \cos n\theta \cos\omega t \\
u_\theta &= A_\theta \sin\xi s \sin n\theta \cos\omega t \\
u_r &= A_r \sin\xi s \cos n\theta \cos\omega t
\end{aligned}
\tag{9}
$$

where A_z, A_θ, A_r, and ξ are undetermined constants, n is an integer greater than or equal to zero, and ω is an angular frequency. Also, the periodic functions of θ in Eq. (9) guarantee that the displacements are periodic and continuous in the circumferential direction. Substituting Eq. (9) into Eq. (5), the following homogeneous equation is derived:

$$
([\mathcal{M}_{\mathrm{D-M}}] + k[\mathcal{M}_{\mathrm{MOD}}])\begin{bmatrix} A_z \\ A_\theta \\ A_r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}
\tag{10}
$$

where

$$
\begin{aligned}
[\mathcal{M}_{\mathrm{D-M}}]_{11} &= -\xi^2 - \frac{(1-\nu)}{2}n^2 + \frac{\rho_{\mathrm{all}}(1-\nu^2)R^2\omega^2}{E} \\
[\mathcal{M}_{\mathrm{D-M}}]_{22} &= -\frac{(1-\nu)}{2}\xi^2 - n^2 + \frac{\rho_{\mathrm{all}}(1-\nu^2)R^2\omega^2}{E} \\
[\mathcal{M}_{\mathrm{D-M}}]_{33} &= 1 + k(\xi^2+n^2)^2 - \frac{\rho_{\mathrm{all}}(1-\nu^2)R^2\omega^2}{E} \\
[\mathcal{M}_{\mathrm{D-M}}]_{12} &= [\mathcal{M}_{\mathrm{D-M}}]_{21} = \frac{(1+\nu)}{2}\xi n \\
[\mathcal{M}_{\mathrm{D-M}}]_{13} &= -[\mathcal{M}_{\mathrm{D-M}}]_{31} = \nu\xi \\
[\mathcal{M}_{\mathrm{D-M}}]_{23} &= -[\mathcal{M}_{\mathrm{D-M}}]_{32} = -n
\end{aligned}
\tag{11}
$$

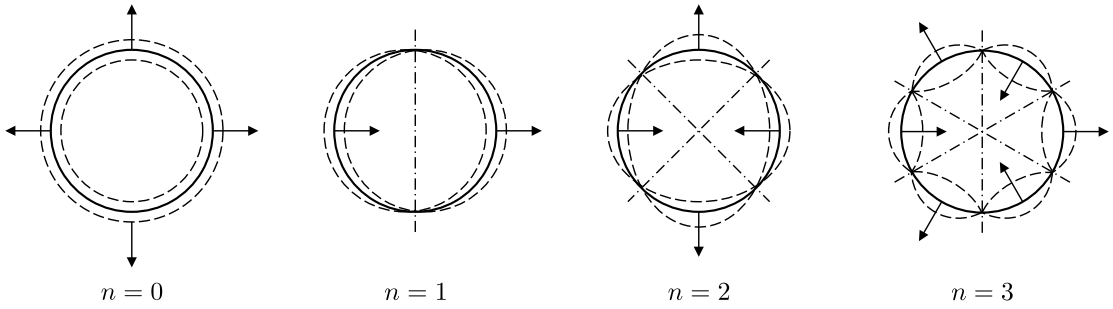


Fig. 2. Vibration modes of the cylinder in the circumferential direction.

and

$$\begin{aligned}
[\mathcal{M}_{\mathrm{MOD}}]_{11} &= -\frac{(1-\nu)}{2}n^2 \\
[\mathcal{M}_{\mathrm{MOD}}]_{22} &= -\frac{3(1-\nu)}{2}\xi^2 \\
[\mathcal{M}_{\mathrm{MOD}}]_{33} &= 1-2n^2 \\
[\mathcal{M}_{\mathrm{MOD}}]_{12} &= [\mathcal{M}_{\mathrm{MOD}}]_{21} = 0 \\
[\mathcal{M}_{\mathrm{MOD}}]_{13} &= -[\mathcal{M}_{\mathrm{MOD}}]_{31} = \xi^3 - \frac{(1-\nu)}{2}\xi n^2 \\
[\mathcal{M}_{\mathrm{MOD}}]_{23} &= -[\mathcal{M}_{\mathrm{MOD}}]_{32} = -\frac{(3-\nu)}{2}\xi^2 n
\end{aligned} \tag{12}$$

For Eq. (10) having a nontrivial solution other than $A_z = A_\theta = A_r = 0$, the determinant of the coefficient matrix must be zero [16]:

$$\det([\mathcal{M}_{\mathrm{D-M}}] + k[\mathcal{M}_{\mathrm{MOD}}]) = 0 \tag{13}$$

The Flügge's shell theory neglects k with respect to unity since it covers thin-walled cylinders [10]. Through solving Eq. (13), a cubic polynomial for the natural frequencies of the cylindrical vibration is derived as follows [10]:

$$\lambda^6 - (K_2 + k\Delta K_2)\lambda^4 + (K_1 + k\Delta K_1)\lambda^2 - (K_0 + k\Delta K_0) = 0 \tag{14}$$

where

$$\lambda^2 = \frac{\rho_{\mathrm{all}}(1-\nu^2){d_{\mathrm{mean}}}^2\omega^2}{4E}, \qquad k = \frac{h^2}{3{d_{\mathrm{mean}}}^2}, \qquad \xi = k_z\frac{d_{\mathrm{mean}}}{2} \tag{15}$$

and

$$\begin{aligned}
K_2 &= 1 + \frac{1}{2}(3-\nu)(n^2+\xi^2) + k(n^2+\xi^2)^2 \\
K_1 &= \frac{1}{2}(1-\nu)\left\{(3+2\nu)\xi^2 + n^2 + (n^2+\xi^2)^2 + \frac{3-\nu}{1-\nu}k(n^2+\xi^2)^3\right\} \\
K_0 &= \frac{1}{2}(1-\nu)\left\{(1-\nu^2)\xi^4 + k(n^2+\xi^2)^4\right\}
\end{aligned} \tag{16}$$

Also, n corresponds to a circumferential vibration mode, and k_z is an axial wave number. K_0, K_1, and K_2 are constants based on the Donnell–Mushtari's shell theory, and ΔK_0, ΔK_1, and ΔK_2 are modifying constants. ΔK_0, ΔK_1, and ΔK_2 for the Flügge's shell theory are described as follows:

$$\begin{aligned}
\Delta K_2 &= 0 \\
\Delta K_1 &= 0 \\
\Delta K_0 &= \frac{1}{2}(1-\nu)\left\{2(2-\nu)\xi^2 n^2 + n^4 - 2\nu\xi^6 - 6\xi^4 n^2 - 2(4-\nu)\xi^2 n^4 - 2n^6\right\}
\end{aligned} \tag{17}$$

Eq. (14) becomes a sixth order equation of the angular frequency ω for fixed values n and k_z, and a continuous solution in the circumferential direction is obtained for each fixed value n as shown in Fig. 2. Also, Eq. (14) has three real solutions for ω, and these are classified as angular frequencies related to the axial, circumferential, and radial motions. The lowest angular frequency is usually associated with the radial motion. Through the process mentioned above, dispersion relations of cylinder walls between k_z and ω are obtained in each n.

Table 1
Material parameters for the calculation of dispersion relations.

Parameter	Description	Reference
$E = 193$ GPa	Young's modulus of SUS304	[26]
$\nu = 0.318$	Poisson's ratio of SUS304	[26]
$\rho_c = 8000$ kg/m^3	Density of SUS304	[26]
$c = 1494$ m/s	Speed of sound in water at 24 °C	[27]
$\rho_f = 997.2$ kg/m^3	Density of water at 24 °C	[28]

2.3. Dispersion relations of internal fluid

To describe the sound propagation of fluid inside a cylinder, we consider the wave equation as expressed in the following equation:

$$\nabla^2\phi(\boldsymbol{r},t) = \frac{1}{c^2}\frac{\partial^2\phi(\boldsymbol{r},t)}{\partial t^2} \tag{18}$$

where c and $\phi(\boldsymbol{r},t)$ denote the speed of sound and the velocity potential of the sound field respectively, and ∇^2 is expressed as follows in the cylindrical coordinates:

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2} \tag{19}$$

Suppose $\phi(\boldsymbol{r},t)$ is separable as $\phi(\boldsymbol{r},t) = \phi(\boldsymbol{r})e^{-i\omega t}$, the Helmholtz equation is derived as follows:

$$\left\{\nabla^2 + \left(\frac{\omega}{c}\right)^2\right\}\phi(\boldsymbol{r}) = 0, \qquad \frac{\partial\phi}{\partial\boldsymbol{n}} = 0 \tag{20}$$

where $\boldsymbol{n}$ denotes the outward normal vector at the boundary, and the right-hand side relation of Eq. (20) corresponds to the boundary conditions. Through variable separation in the axial, circumferential, and radial directions, the solution of $\phi(z,\theta,r,t)$ is derived as follows [25]:

$$\phi(z,\theta,r,t) = \sum_{n,m,l} A_{nml}\cdot J_n\left(\frac{u_{nm}}{d_{\mathrm{in}}/2}r\right)\cdot\cos n\theta\cdot\cos k_z z\cdot\cos(\omega t - \alpha_{nml}) \tag{21}$$

where J_n is the Bessel function of the first kind as shown in the following equation:

$$J_n(x) = \sum_{m=0}^{\infty}\frac{(-1)^m}{m!\Gamma(n+m+1)}\left(\frac{x}{2}\right)^{n+2m} \tag{22}$$

and u_{nm} is the mth solution of $\partial J_n/\partial x = 0$. Also, A_{nml} and α_{nml} are constants, each of which represents a coefficient related to an amplitude and a phase angle. From Eq. (21), sound pressure distributions in the cylindrical cross-section are represented as shown in Fig. 3, in which n and m correspond to the circumferential and radial vibration modes, respectively. Also, solid black lines represent the boundaries where the sound pressure is greater than zero or not, and the sound pressure distribution is divided into $2n$ areas in the circumferential direction. In addition, a resonant angular frequency of the internal fluid in the cylinder is derived as shown in the following equation:

$$\omega = c\sqrt{\left(\frac{u_{nm}}{d_{\mathrm{in}}/2}\right)^2 + {k_z}^2} \tag{23}$$

Through calculating ω in Eq. (23), dispersion relations of the internal fluid between k_z and ω are obtained.

2.4. Coupled waves of cylinder walls and internal fluid

In the present study, we employ water flow inside a pipe made of SUS304 with $d_{\mathrm{in}} = 158.2$ mm and $d_{\mathrm{out}} = 165.2$ mm, and calculate dispersion relations of the cylinder walls and internal fluid using material parameters of SUS304 and water shown in Table 1. Fig. 4 represents the calculation results of the cases of $n = 0,\ 1,\ 2,\ 3,\ 4$, and 5, in which blue and orange lines show dispersion relations of the cylinder walls and internal fluid, respectively. First, looking at the dispersion relations of cylinder walls in the cases of $n = 0,\ 1,\ 2$, and 3, Eq. (14) has all solutions of $f_i = \omega_i/2\pi$ $(i = 1,\ 2,\ 3)$ in the range from 0 to 25 kHz. However, in the cases of $n = 4$ and 5, the biggest solution of Eq. (14) is greater than 25 kHz, that is, there are two solutions of $f_i = \omega_i/2\pi$ $(i = 1,\ 2)$ in the range from 0 to 25 kHz. Also, the frequency and wave number indicate the dispersibility of the non-linear relation except for one mode in the case of $n = 0$ (see the * line in Fig. 4). Now for the dispersion relations of the internal fluid, all vibration modes except $n = m = 0$ have the cut-off frequency of $k_z = 0$, in which the phase velocity is infinite. Below the cut-off frequency, the sound wave rarely propagates because k_z becomes the complex number and the sound wave in each mode decreases exponentially over the axial distance.

Next, we discuss the vibration forms at intersection points of dispersion relations of the cylinder walls and internal fluid in Fig. 4. At these intersection points, dispersion relations of the cylinder walls and internal fluid have identical axial wave numbers and identical frequencies. In each of these cases, the axial transverse wave of the cylindrical vibration has an identical wavelength

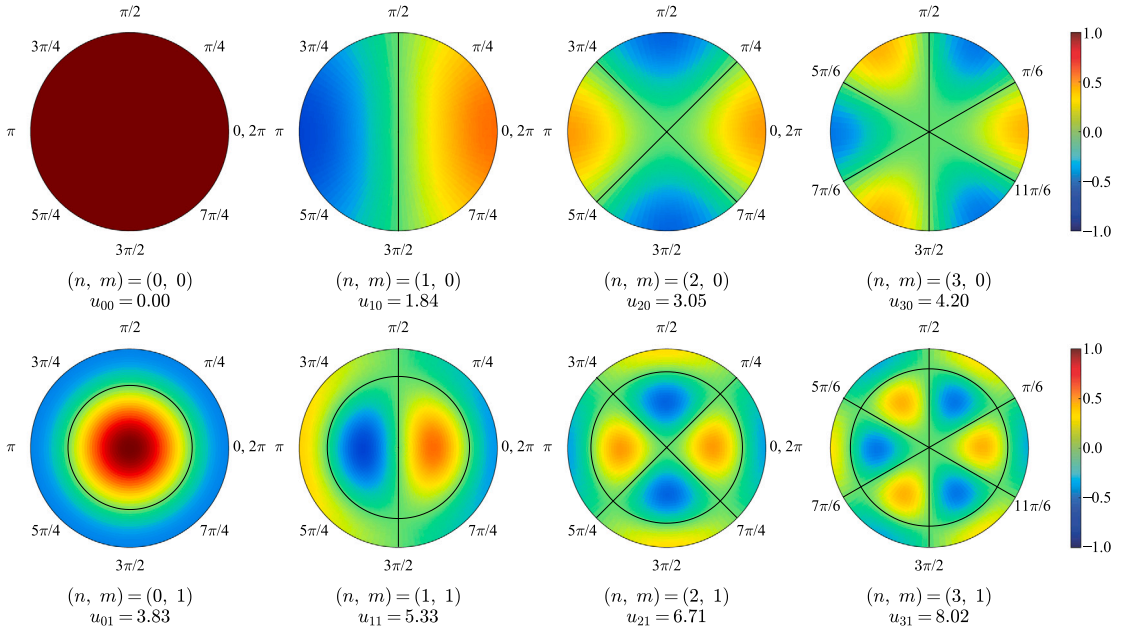


Fig. 3. Sound pressure distribution in the cylindrical cross-section derived by $J_n(u_{nm}r/(d_{\text{in}}/2)) \cdot \cos n\theta$.

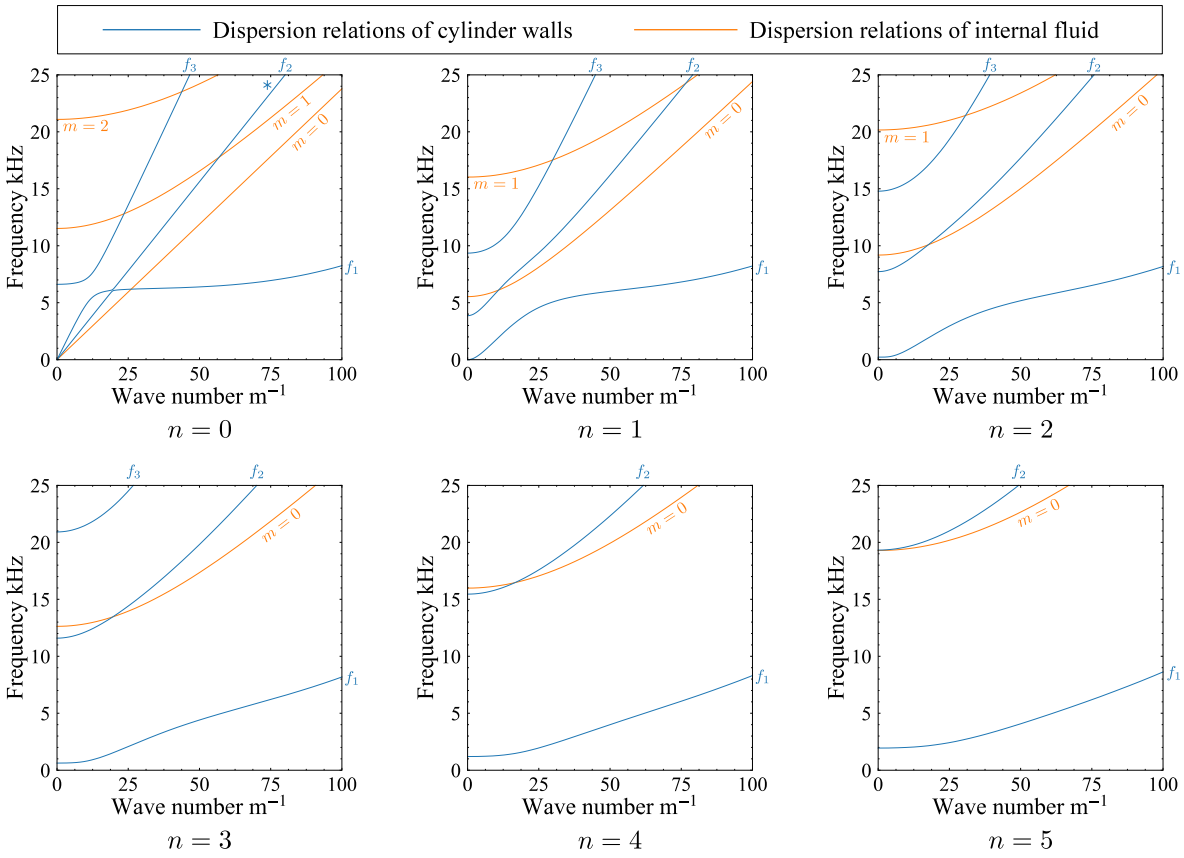
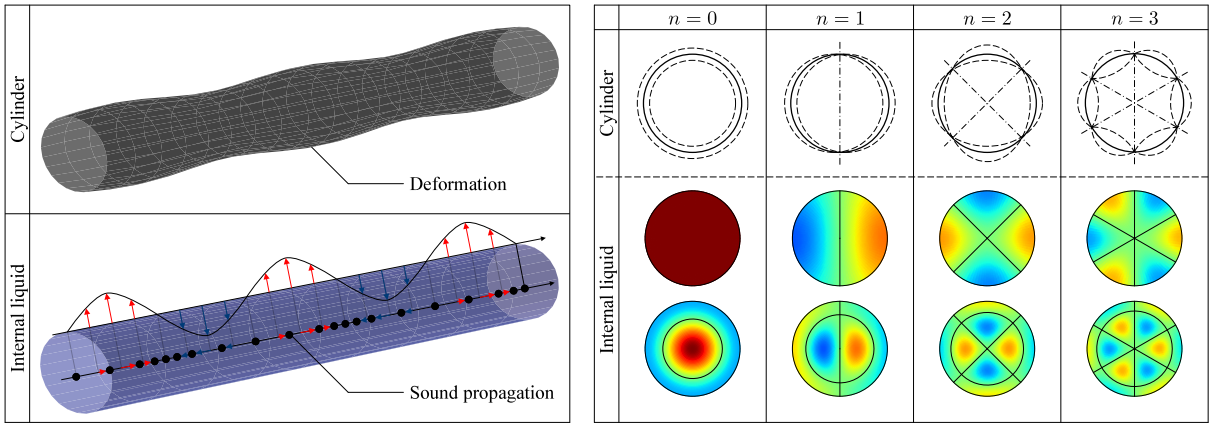


Fig. 4. Dispersion relations of the cylinder walls and internal fluid for n = 0, 1, 2, 3, 4, and 5. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Axial direction (b) Circumferential direction

Fig. 5. Coupling of the cylindrical vibration and sound propagation of the internal fluid.

to the axial longitudinal wave of the sound propagation of the internal fluid at an identical frequency as shown in Fig. 5(a). Therefore, the cylindrical vibration and sound propagation of the internal fluid are coupled in the axial direction. Moreover, in each circumferential vibration mode n, the cylindrical vibration has $2n$ nodes and the sound pressure distribution is divided into $2n$ areas in the circumferential direction as shown in Fig. 5(b). Hence, the cylindrical vibration and the sound propagation of the internal fluid are also coupled in the circumferential direction. Considering these things, the cylindrical vibration and the sound propagation of the internal fluid can cause the dominant coupled waves at the frequencies of intersection points. As shown in Fig. 4, coupled wave frequencies exist in the range from 0 to 25 kHz in the cases of $n = 0,\ 1,\ 2,\ 3$, and 4. However, they do not exist in this range in the cases of $n \geq 5$. These coupled wave frequencies can be determined by a numerical method. In the present study, we first calculate the dispersion relations of cylinder walls (from Eq. (14)) and internal fluid (from Eq. (23)) as shown in Fig. 4. Second, the intersection points are determined as those where the difference between the dispersion relations is numerically minimized. The frequencies at these intersection points correspond to the theoretical values of coupled wave frequencies.

3. Experimental arrangement

To validate the theory mentioned in Chapter 2, we conduct an experiment to observe coupled waves of cylinder walls and an internal fluid. The experimental arrangement is shown in Fig. 6. A pipe made of SUS304 is supported horizontally and its inner and outer diameters are $d_{\mathrm{in}} = 158.2$ mm and $d_{\mathrm{out}} = 165.2$ mm, respectively. Also, the working fluid inside the pipe is pure water. These conditions are consistent with the calculation conditions of dispersion relations in Chapter 2. To measure the sound and vibration of pipe walls and internal water, we use a hydrophone (ACO TYPE7800, Japan), which has the flat frequency response and the omni-directional. This hydrophone is located 200 mm below the lower end of the pipe and about $15d_{\mathrm{in}}$ downstream from the pipe inlet in a water tank with 300 mm in width, 450 mm in depth, and 600 mm in height. In this experiment, the sound and vibration are caused by the pipe and water and there are not any external vibration sources. We measure the sound and vibration in the cases of flow rates of 0, 10, 30, 60, 80, 100, 125, 150, 180, 210, 225, and 240 m^3/h at a sampling rate of 44.1 kHz for 1 min.

4. Results and discussion

The present study performs frequency analysis based on the PSD using Welch's method [29]. First, discrete original signals $x[1],\ x[2],\ \ldots,\ x[N]$ are divided into L segments with a length of M data points while S data points are shifted between segments. For each segment, the lth windowed discrete Fourier transform is calculated as follows:

$$X_l(f) = \sum_{m=(l-1)S+1}^{M+(l-1)S} x[m]w[m]\exp(-2\pi \mathrm{j} f m) \qquad (l = 1,\ \ldots,\ L) \tag{24}$$

where w is a window function, and f is a frequency. The PSD is computed as shown in the following equation:

$$S(f) = \frac{1}{FLW}\sum_{l=1}^{L}\left|X_l(f)\right|^2 \tag{25}$$

where F is a sampling rate (= 44.1 kHz), and W denotes a power parameter factor of the window function defined as:

$$W = \sum_{m=1}^{M} w^2[m] \tag{26}$$

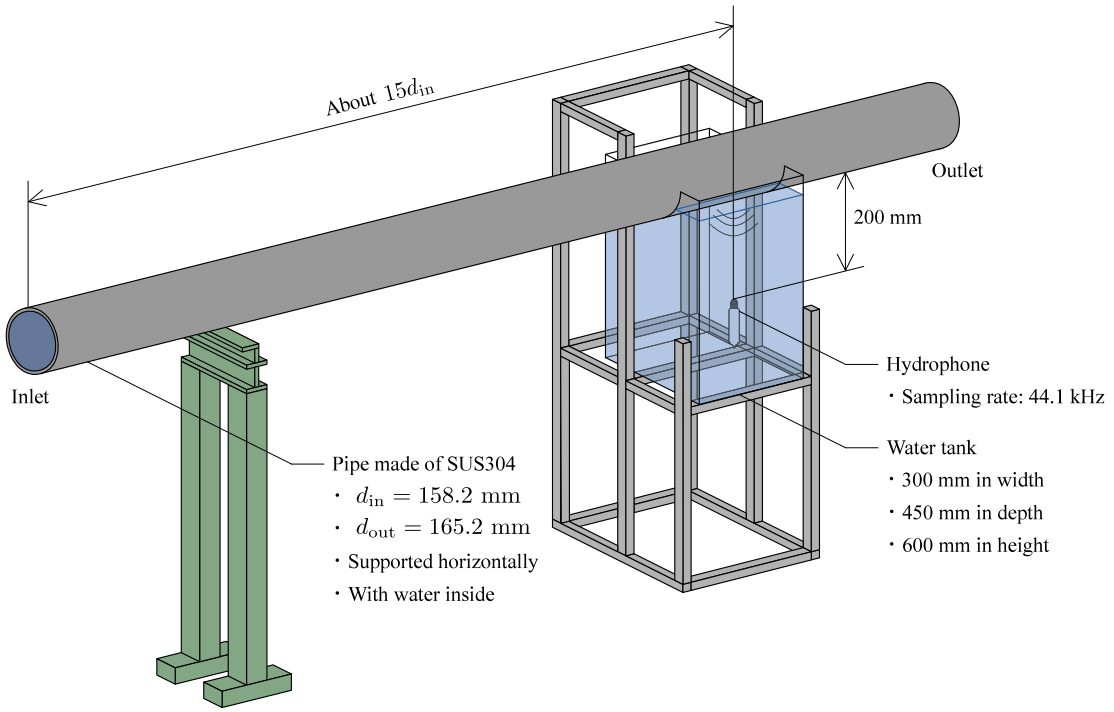


Fig. 6. Experimental arrangement of sound and vibration acquisition.

Also, if a series of discrete original signals has the Gaussian distribution, the PSD will have the χ^2 distribution with degrees of freedom $2L$ [30]. Through this theory, the 100α% confidence interval of the PSD is calculated as follows [31]:

$$\frac{\nu_{\mathrm{DOF}} \cdot S(f)}{\chi^2(\nu_{\mathrm{DOF}}, (1+\alpha)/2)} \leq S(f) \leq \frac{\nu_{\mathrm{DOF}} \cdot S(f)}{\chi^2(\nu_{\mathrm{DOF}}, (1-\alpha)/2)} \tag{27}$$

where ν_{DOF} denotes degrees of freedom ($\nu_{\mathrm{DOF}} = 2L$), and $\chi^2(\nu_{\mathrm{DOF}}, (1\pm\alpha)/2)$ represents the $100\times(1\pm\alpha)/2$% point of the χ^2 distribution with degrees of freedom $\nu_{\mathrm{DOF}} = 2L$.

The PSD of measured signals is computed through the process described above. First, noise in the low frequency band is removed by a high-pass filter with a stopband of 900 Hz or less and a passband of 1000 Hz or more. Also, we employ the Hanning window as a window function, and set the parameters of Welch's method as follows: $L = 1000$, $M = 4410$, and $S = 2205$. In this condition, discrete signals for 1 min in each experimental condition are divided into 1000 segments with a length of 4410 data points while 2205 data points are shifted between segments; each segment has data points for 0.1 s, and an overlap rate is set to be 50%. Fig. 7 shows the PSD of coupled waves of the cylinder walls and internal fluid measured in the experiment. In Fig. 7, blue solid lines represent the PSD, and the shadings of the same thin color express 95% confidence intervals of the PSD ($\alpha = 0.95$). To confirm the existence of the confidence intervals of the PSD, we show an enlarged view around a peak in the case of 80 m^3/h in Fig. 7. This enlarged view demonstrates that the confidence intervals of the PSD remain within a small range because of a large number of segments, and the same applies to other conditions. The following is the analysis in terms of the generation of coupled waves, vibration modes, and coupled wave frequencies.

4.1. Generation of coupled waves

In the experiment, coupled waves are observed even without the vibration source. This is because the viscosity and density of liquid are larger than those of gas, which will intensify the flow-induced vibration. Also, coupled waves are observed in the cases of 80, 100, 125, 150, 180, 210, and 240 m^3/h. The power of coupled waves becomes the largest in the case of 150 m^3/h. On the other hand, coupled waves are not observed despite the high flow rate such as 225 m^3/h. This result demonstrates that the flow rate and the power of coupled waves have a non-linear relationship.

4.2. Vibration mode

Vibration modes observed in each condition are shown in Table 2. The rightmost column shows the peak numbers of the PSD, which correspond to the arrow numbers in Fig. 7. As shown in Table 2, vibration modes of $n = 0,\ 1,\ 2,\ 3$ and 4 are observed in the experiment, and the occurring vibration modes and the most prominent vibration mode vary with the flow rates:

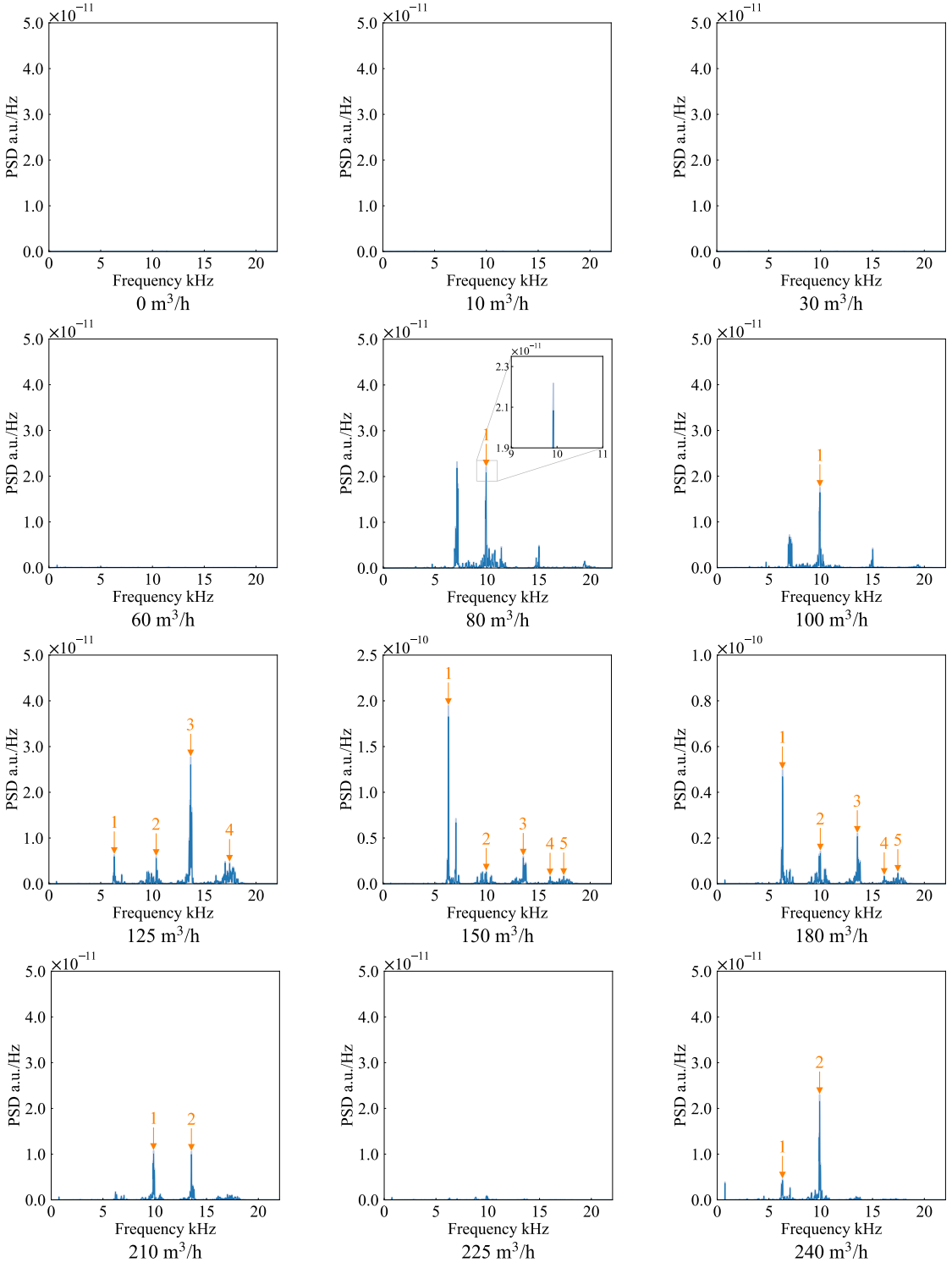


Fig. 7. Power spectral density of coupled waves of the cylinder walls and internal fluid. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Comparison between theoretical and experimental coupled wave frequencies.

Flow rate m^3/h	Reynolds number	Vibration mode (n, m)	Coupled wave frequency kHz		Peak No.
			Theoretical value	Experimental value	
0 ~ 60	0 ~ 1.5×10^5	–	–	–	–
80	2.0×10^5	(2, 0)	10.09	9.92	1
100	2.6×10^5	(2, 0)	10.09	9.92	1
125	3.2×10^5	(0, 0)	6.19	6.31	1
		(2, 0)	10.09	10.36	2
		(3, 0)	13.46	13.69	3
		(1, 1)	17.53	17.02	4
150	3.8×10^5	(0, 0)	6.19	6.31	1
		(2, 0)	10.09	9.97	2
		(3, 0)	13.46	13.54	3
		(4, 0)	16.47	16.13	4
		(1, 1)	17.53	17.45	5
180	4.6×10^5	(0, 0)	6.19	6.31	1
		(2, 0)	10.09	9.97	2
		(3, 0)	13.46	13.54	3
		(4, 0)	16.47	16.13	4
		(1, 1)	17.53	17.45	5
210	5.4×10^5	(2, 0)	10.09	9.89	1
		(3, 0)	13.46	13.54	2
225	5.7×10^5	–	–	–	–
240	6.1×10^5	(0, 0)	6.19	6.31	1
		(2, 0)	10.09	9.89	2

- In the cases of 80 and 100 m^3/h, only the vibration mode of $(n,\ m) = (2,\ 0)$ is observed.
- In the case of 125 m^3/h, the vibration modes of $(n,\ m) = (0,\ 0)$, $(2,\ 0)$, $(3,\ 0)$, and $(1,\ 1)$ are observed, and the power is the largest for $(n,\ m) = (3,\ 0)$.
- In the cases of 150 and 180 m^3/h, the vibration modes of $(n,\ m) = (0,\ 0)$, $(2,\ 0)$, $(3,\ 0)$, $(4,\ 0)$, and $(1,\ 1)$ are observed, and the power is the largest for $(n,\ m) = (0,\ 0)$.
- In the case of 210 m^3/h, the vibration modes of $(n,\ m) = (2,\ 0)$ and $(3,\ 0)$ are observed, and the power is the largest for $(n,\ m) = (2,\ 0)$.
- In the case of 240 m^3/h, the vibration modes of $(n,\ m) = (0,\ 0)$ and $(2,\ 0)$ are observed, and the power is the largest for $(n,\ m) = (2,\ 0)$.

4.3. Coupled wave frequencies

The theoretical and experimental values of coupled wave frequencies observed in each condition are shown in Table 2. Looking at the coupled wave frequencies, the theoretical values agree consistently with the experimental values in each vibration mode shown in the above bulleted list. This result indicates that coupled wave frequencies can be predicted correctly by calculating frequencies at intersection points of dispersion relations of cylinder walls and an internal fluid with the consideration of the fluid added mass. We speculate that the differences between theoretical and experimental values are caused by the material parameters used in the calculation of dispersion relations or the experimental condition that outside of the pipe contacts water in the water tank, which is different from the calculation condition that the vibration behavior is solved as that in the air only considering the fluid added mass of internal water.

In addition, there are peaks other than those caused by the coupled waves. Among them, the peak in the lower frequency band (up to 1 kHz) in the case of 240 m^3/h can be described by the natural vibration of the pipe. On the other hand, peaks in the higher frequency band (more than 5 kHz) in the cases of 80 and 100 m^3/h are caused by the standing waves in the cylinder or the increase of sound pressure in the pipe as close to the cut-off frequency of each vibration mode [9,32].

5. Conclusion

The present study has investigated the coupled waves of cylinder walls and an internal liquid and proposed a novel prediction method of coupled wave frequencies of cylinders with liquid inside. First, we have newly introduced the theory of the fluid added mass into the cylindrical shell theory, and calculated the dispersion relations of the cylinder walls and internal liquid. Second, we have conducted the experiment to validate our proposed theory. Finally, we have analyzed the observed coupled waves and compared the experimental and theoretical values of coupled wave frequencies based on the PSD. Our findings are summarized as follows:

1. The coupled waves occur even without any external vibration sources in the case of water flow in the pipe. The generation and power of coupled waves have a non-linear relationship with the flow rate, and coupled waves may not be observed despite the high flow rate.
2. When coupled waves occur, the vibration modes and the most prominent vibration mode vary with the flow rate.
3. The experimental values of coupled wave frequencies agree consistently with the theoretical values calculated by dispersion relations of the cylinder walls and internal fluid. This result demonstrates that the coupled wave frequencies can be predicted correctly using dispersion relations derived from the shell theory considering the fluid added mass and the wave equation. The prediction of coupled wave frequencies can avoid the significant levels of vibrations and noises by making a difference between coupled wave frequencies and mechanical vibration frequencies.

We have demonstrated the consistent agreement between experimental and theoretical values of coupled wave frequencies, and we have also predicted possible excited vibration modes in each case. In future works, we will comparably investigate experimental and theoretical values of vibration modes through an experimental modal analysis, which leads to an improvement in the validity of our proposed method. Although we have observed the variation of excited vibration modes with the flow rates or the corresponding Reynolds numbers, the actually excited vibration modes in each condition are not identified without experiments. Experiments using cylinders made of different materials or different types of internal fluids have the possibility to clarify the relationships between excited vibration modes and flow rates.

CRediT authorship contribution statement

Nao Mikami: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Yoshitaka Ueki:** Conceptualization, Methodology, Investigation, Resources, Writing – review & editing, Project administration, Funding acquisition. **Masahiko Shibahara:** Methodology, Resources, Supervision. **Kosuke Aizawa:** Conceptualization, Methodology, Investigation, Resources, Project administration, Funding acquisition. **Kuniaki Ara:** Conceptualization, Methodology, Supervision.

Declaration of competing interest

This manuscript has not been published and is not under consideration for publication elsewhere. We have no conflicts of interest to disclose.

On behalf of my co-authors, I greatly appreciate your consideration.

Data availability

Data will be made available on request.

Acknowledgments

The paper includes the results of “Technical development program on a common base for fast reactors” entrusted to JAEA by Ministry of Economy, Trade and Industry. This work was supported by JSPS, Japan KAKENHI Grant Number JP23K03713.

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